

Synergistic co-steam gasification of biomass and refuse-derived fuel: A path to enhanced gasification performance

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Abstract: The co-steam gasification of biomass (straw) and Refuse-Derived Fuel (RDF) presents a promising pathway for sustainable waste management and renewable energy production, with significant implications for environmental protection. This study investigates the impact of gasification temperature and steam-to-biomass (S/M) ratio on the process's efficiency, syngas quality, and environmental outcomes. The experiments reveal that optimizing these operational parameters is crucial for enhancing syngas production-rich in hydrogen (H_2) and carbon monoxide (CO)-while minimizing the generation of tar and other undesirable by-products. An optimal S/M ratio was identified that maximizes H_2 and CO yields and promotes tar conversion efficiency, thereby improving the environmental profile of the gasification process. Higher gasification temperatures were found to further increase syngas yield and quality by facilitating mineral transformations and structural changes in the solid residue, thus supporting energy recovery and minimizing environmental hazards. Notably, the co-gasification of RDF and straw exhibited synergistic effects, leading to enhanced syngas yields, improved carbon conversion, and cold gas efficiencies, especially at elevated temperatures and optimal S/M ratios. These synergies, attributed to catalytic interactions between feedstock constituents, result in a gas output with reduced CO_2 and CH_4 levels, indicative of an optimized energy source. The study underscores the potential of co-steam gasification of straw and RDF as a technologically viable and environmentally friendly approach to waste-to-energy conversion, emphasizing the importance of operational optimization for achieving superior energy recovery, resource efficiency, and reduced environmental impact.

Keywords: Municipal solid waste; biomass; gasification; syngas; synergistic enhancement.

1. Introduction

As the world strides toward a more sustainable energy future, the integration of renewable energy sources has become imperative. Among these, biomass stands out as a prominent, carbon-neutral resource due to its abundant availability and renewable nature [1]. However, biomass utilization is often hindered by its intrinsic properties, such as low energy density and heterogeneity [2]. Parallel to this, the escalating volumes of municipal solid waste (MSW) have given rise to the generation of refuse-derived fuel (RDF), which encompasses a high-energy-value waste fraction that can be recovered as energy instead of being landfilled [3]. While RDF offers a significant energy recovery option, its gasification is often challenged by the presence of diverse and complex materials [4]. The steam co-gasification of biomass and RDF presents a viable solution that harnesses the strengths of both feedstocks while mitigating their individual limitations through synergistic interactions.

Synergy in co-gasification processes arises when the interactions between biomass and RDF during gasification lead to improved gas yields, calorific values, and carbon conversion efficiency, as compared to their individual gasification [5]. These benefits are attributed to the complementary properties of the feedstocks. For instance, RDF often contains plastics and other materials with high calorific value that can increase the energy content of the gas product [6]. In contrast, biomass can contribute to the stability of the process with its consistent composition and char-forming ability, which can enhance tar cracking and improve syngas quality [7].

The influence of gasification temperature on the co-gasification of biomass and RDF is a subject of considerable study. Higher temperatures typically enhance the gasification reactions, leading to increased syngas yields and improved carbon conversion efficiencies [8]. However, the optimal temperature range must be identified to maximize synergetic effects while avoiding excessive thermal degradation of the feedstocks, which could lead to the formation of undesirable byproducts. The S/M ratio plays a pivotal role in the gasification process, influencing the steam reforming of hydrocarbons, water-gas shift reactions, and the overall energy balance of the system [9]. A carefully optimized S/M ratio can promote the production of hydrogen, an essential component of high-quality syngas, and reduce the formation of tars and chars [9]. The determination of the ideal S/M ratio is thus crucial for achieving optimal syngas composition and maximizing the synergetic effects between biomass and

RDF. Analyzing the gasification characteristics involves assessing the composition and calorific value of the syngas produced, as well as the cold gas efficiency, which provides an indication of the overall energy efficiency of the gasification process. Furthermore, the carbon conversion efficiency is a critical parameter, reflecting the effectiveness of the gasification in transforming feedstock carbon into gas-phase products [10]. The characterization of the residual solids from gasification, including their composition and morphology, is equally important. Advanced characterization techniques such as Energy Dispersive Spectroscopy (EDS) and Scanning Electron Microscopy (SEM) are employed to analyze the solid byproducts. EDS provides elemental composition data, which can inform the understanding of ash behavior and the potential for catalyst activity within the residual solids. SEM offers insights into the physical structure and the morphology of the chars, which can be correlated with reactivity and gasification dynamics [11].

Therefore, the synergy in co-gasification is not just a theoretical concept but also a practical approach to optimize the energy output from available resources. The combined gasification characteristics, such as syngas composition, thermal value, cold gas efficiency, and carbon conversion efficiency, provide a comprehensive picture of the process performance. Moreover, the physicochemical analysis of the gasification residues through EDS and SEM further elucidates the transformation mechanisms during co-gasification, shedding light on the intricate interplays at work.

This study aims to systematically investigate the individual and combined effects of gasification temperature and S/M ratio on the co-gasification of biomass and RDF. By analyzing the process outputs and residue characteristics, we seek to optimize the gasification parameters for enhanced synergetic effects, leading to the production of high-quality syngas and the efficient utilization of solid residues. The outcomes of this research will have implications for the design and operation of co-gasification systems, potentially contributing to a more sustainable and effective waste-to-energy conversion technology.

2. Experimental

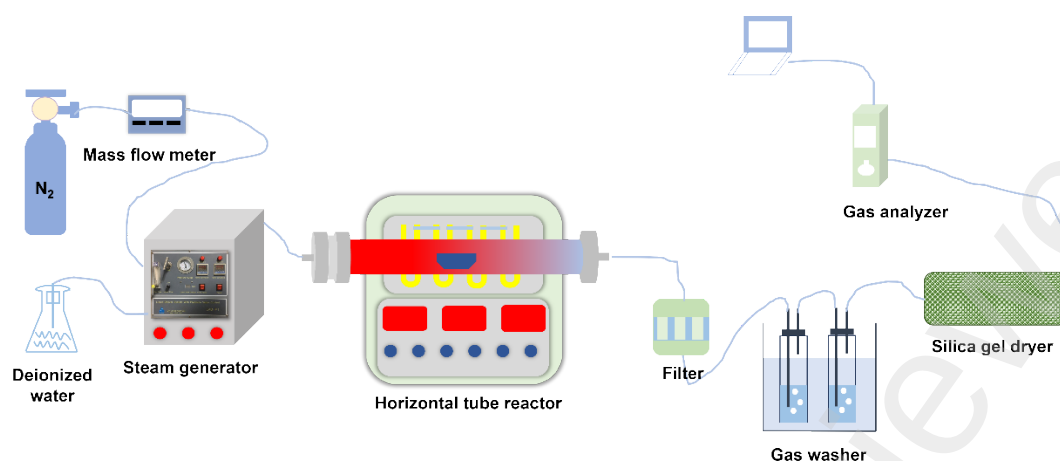
2.1. Characteristics of biomass and RDF

The straw and refuse-derived fuel (RDF) utilized in the experimental procedures of this study have been sourced consistent with the feedstock characteristics reported in our team's prior

publications [12]. Detailed compositional and proximate analyses, alongside industrial characterizations of the biomass and RDF, have been extensively documented in these previous works. For a comprehensive understanding of the feedstock properties, readers are referred to these foundational studies [12]. This established baseline of knowledge has been instrumental in informing the experimental design and facilitating a deeper interpretation of the gasification characteristics observed in the present study.

2.2. Examples of biomass and RDF steam gasification

Investigations into the co-steam gasification of RDF and straw were conducted using a fixed-bed laboratory reactor, as depicted in **Scheme 1**. The procedure commenced with nitrogen at a flow rate of 3L/min, constituting 5% of the total flow to serve as the carrier gas for the steam. Upon reaching the pre-set temperature of 220°C by the heating belt, the steam generator was activated. The liquid pump was then engaged at various speeds ($S/M=0.25, 0.50, 0.75$, or 1.00) and allowed to stabilize the steam flow. A constant total gas flow rate of 150 mL/min was maintained to negate the influence of flow rate variations on gasification outcomes. Experimental samples weighing 2 g and having a particle size of 0.075 mm, were placed in an alumina crucible at the reactor's entrance. Nitrogen at 150 mL/min was introduced for 20 minutes to purge the reactor of residual air. The reactor's heating rate was set at 15 °C/min until reaching the target temperatures of 600°C, 700°C, 800°C, or 900°C. The crucible was then repositioned to the reactor's center for a 20-minute duration. Syngas was cooled in a water bath to remove tar and moisture. Given that the majority of syngas was produced within the initial 10 minutes, it was captured in sampling bags during this interval. Subsequent gases were combusted. Tar and dry syngas were analyzed using a gas chromatograph (GC model 3420A, Beijing Beifen-Ruili Analytical Instrument Group Co., Ltd). Each sample underwent triplicate testing, with the mean value recorded as the final result.



Scheme 1. Diagram of the laboratory scale fixed-bed reactor facility.

The volume fraction of the syngas (V_i), the volume yield of the syngas composition ($Y_{G, i}$, Nm^3/kg), the LHV of syngas (MJ/kg), the carbon conversion efficiency (CC), and the cold gas efficiency (CGE) were calculated based on the related equations in our previous report [12]. Previous studies indicated that co-gasification processes incorporating 40-50% MSW (or RDF) yield satisfactory results. Consequently, we employed a 1:1 mass ratio of RDF to straw as the feedstock for co-gasification in our study [13-15].

2.3. Characterization

Sample compositions were characterized by X-ray diffraction (XRD) using a D/max2400 instrument (Rigaku Corporation, Japan). Microstructural analysis was conducted with a field emission scanning electron microscope (FE-SEM; Gemini SEM 500, Zeiss, Germany). Elemental analysis was performed on the same FE-SEM model, which was outfitted with an Oxford energy-dispersive spectrometer (EDS).

3. Results and discussion

3.1. Characteristics of biomass gasification

Fig. 1 comprehensively examines the effects of varying steam-to-mass of biomass (S/M) ratios on the gasification characteristics of straw. Fig. 1a manifests the direct relationship between S/M ratio and gas composition. Notably, an optimal S/M ratio seems to maximize H_2 and CO yields, key components of syngas, which is desirable for conversion to liquid fuels or power generation. This is coherent with the enhanced reforming and water-gas shift reactions reported in contemporary studies

focused on biomass gasification [16]. Furthermore, the diminishing outputs of CO_2 , CH_4 , and hydrocarbons with increasing S/M ratios may signal a shift in the reaction pathways towards those favoring syngas production over carbon reformation as described by Shim et al.[5]. Fig. 1b reveals an inverse relationship between the LHV and the S/M ratio, with a concomitant increase in the H_2/CO ratio. This indicated that an increase in steam availability relative to biomass facilitates tar cracking and reforming, thus enriching the hydrogen content in the produced gas.

Fig. 1c relates carbon conversion efficiency and cold gas efficiency to the S/M ratio. The carbon conversion efficiency measures the extent to which carbonaceous matter in the straw is transformed into gaseous constituents [17], whereas the cold gas efficiency provides an indication of the net calorific value of the gas produced relative to the straw input [18]. The non-linear effects of S/M ratios on carbon conversion and cold gas efficiency were depicted. A discernible vertex suggested an optimal S/M ratio, beyond which the further addition of steam leads to a reduction in efficiency, possibly attributable to the saturation of reactive sites or heat consumption by excess steam. Fig. 1d compares the by-product yield distributions, including tar, solids, and gas, across varying S/M ratios of 0.25, 0.5, 0.75, and 1.0. Tar production minimization is a critical target in gasification processes, as high tar levels can lead to downstream equipment fouling. The decreasing trend of tar with increasing S/M ratios was consistent with the hypothesis of enhanced tar cracking under steam-rich conditions, an observation in line with recent studies that advocate for steam-enabled tar reduction strategies [19].

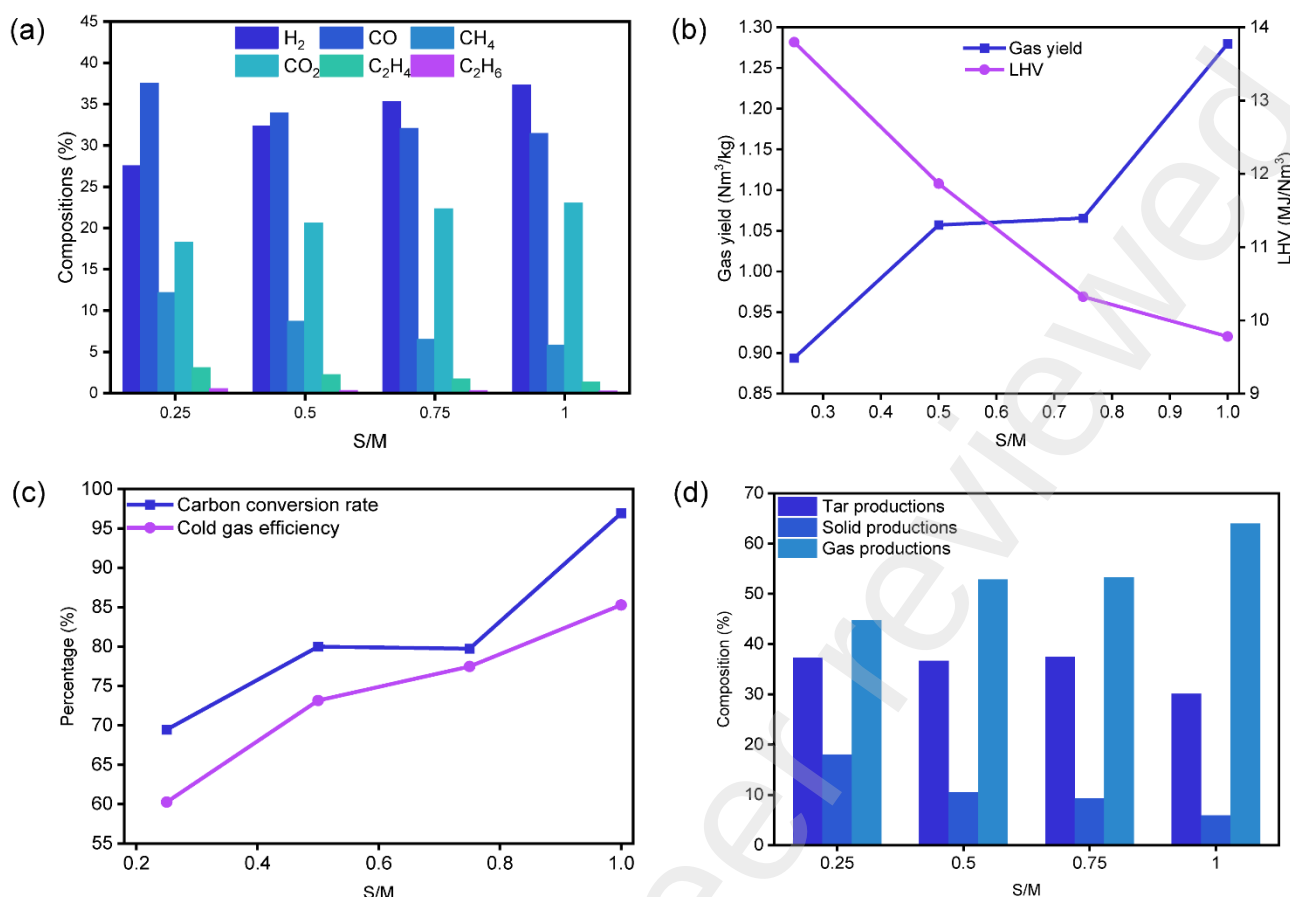


Fig. 1 Effects of S/M on various parameters relevant to the steam gasification of straw: (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions compositions.

Fig. 2a shows the compositions of the primary syngas constituents as a function of temperature. With increasing temperature, clear upward trends for both H₂ and CO concentrations were observed, indicating enhanced gasification reactions and potential for improved syngas quality. The data suggested that higher operational temperatures favored the steam reforming of tars and the water-gas shift reaction, thus boosting H₂ production. A dual-y axis graph demonstrated the variation in gas yield and LHV with temperature (Fig. 2b). Both the increase in gas yield and the raised in LHV with increasing temperatures reaffirm the positive effect of temperature on gas yield and LHV, thereby improving the clean energy potential of the gasification process. The relationship between temperature and efficiency metrics such as carbon conversion and cold gas efficiency was depicted in Fig. 2c. Both efficiencies increased with temperature, indicating a complete conversion of straw into syngas and a higher relative calorific value of the syngas produced. These trends illustrated that operating at elevated temperatures could enhance overall energy recovery from straw gasification.

Fig. 2d compares the yields of tar, solids, and gas at different temperatures. Specifically, it shows the production of gas compared to that of tar and solid production. The quantities of these by-products at varying temperatures help discern the operational conditions that minimize waste and optimize syngas output.

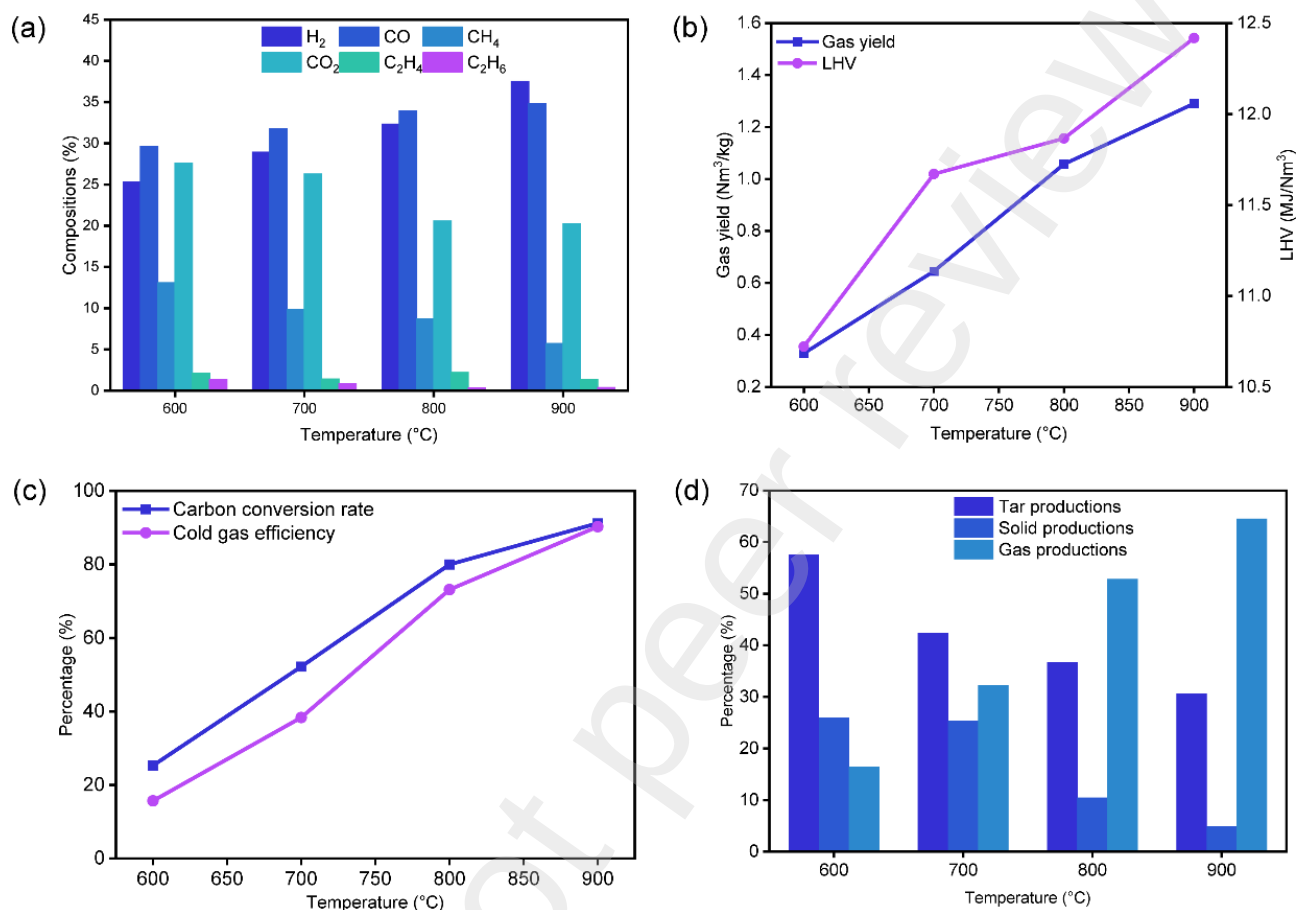


Fig. 2 Effects of temperature on various parameters relevant to the steam gasification of straw: (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions compositions.

Fig. 3 presents the characterization results of solid residues derived from the steam gasification of straw at different temperatures. The XRD patterns of the solid by-product at gasification temperatures of 600°C, 700°C, 800°C, and 900°C provided a showcase of the crystalline phases present within the residue (Fig. 3a). Peaks indicating the presence of potassium chloride (KCl) and potassium sulfate (K₂SO₄) were observed, with their relative intensity varying with temperature. The variation in peak intensity and the emergence or disappearance of certain peaks reflected the thermal transformations occurring within the inorganic components of the straw ash during gasification. It suggested that as the temperature increases, mineral content might transform due to processes like

decomposition, volatilization, and reaction with other ash components.

The EDS analysis provided a semi-quantitative elemental composition of the solid residues (Fig. 3b). The bar chart showed the mass percentage of various elements (C, O, Na, Mg, Al, Si, K, Ca, Fe) identified in the residues at each temperature. Notably, the concentration of certain elements like potassium (K) appeared to change with temperature, which might indicate volatilization or other thermochemical alterations. The relative abundance of other elements such as silicon (Si) and calcium (Ca) also varies, which could be related to the mineral transformations and interactions at different temperatures, as suggested by the XRD results.

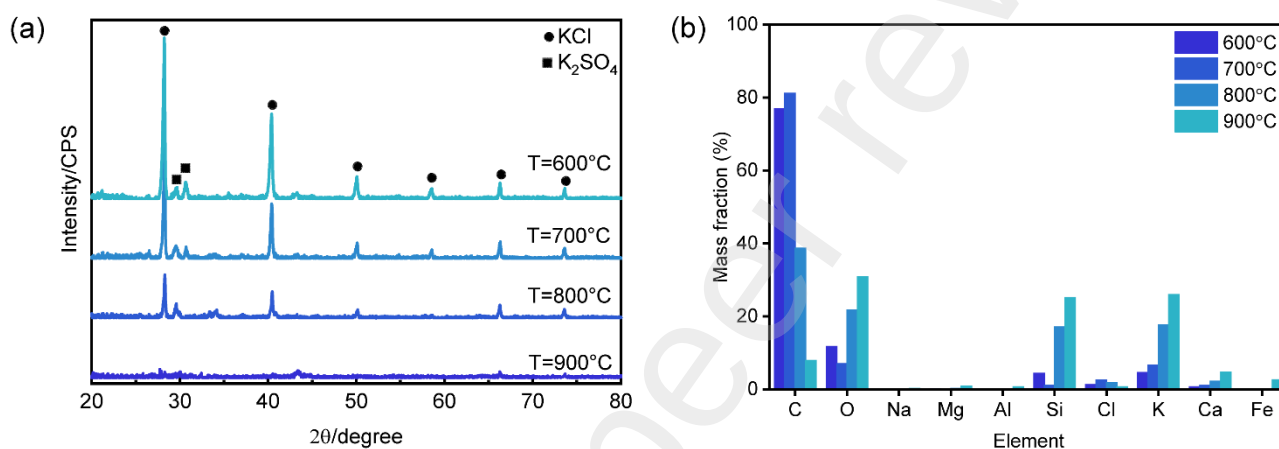


Fig. 3 The characterization results of solid residues derived from the steam gasification of straw at different temperatures: (a) for X-Ray Diffraction (XRD) analysis and (b) for Energy-dispersive X-ray Spectroscopy (EDS) results.

The provided SEM images delineate the progressive structural changes in straw residues at escalating temperatures of 600°C, 700°C, 800°C, and 900°C during gasification, as shown in Fig. 4. At 600°C, the straw exhibits preservation of the elongated fibrous structure with minimal thermal degradation. As the temperature advanced to 700°C, we observed the onset of structural disintegration and increased surface etching, likely indicative of enhanced volatile release. The image at 800°C shows further degradation, with the fibers starting to fragment and visibly distinct pores forming, a suggestion of active char development. At the pinnacle temperature of 900°C, the structure was transformed into a highly porous char, evidencing considerable decomposition and potential catalysis of reaction pathways towards complete gasification. The fibrous texture being conspicuously consumed. This temperature-dependent morphological transformation reflected the thermochemical

reactions facilitating biomass conversion into syngas and char.

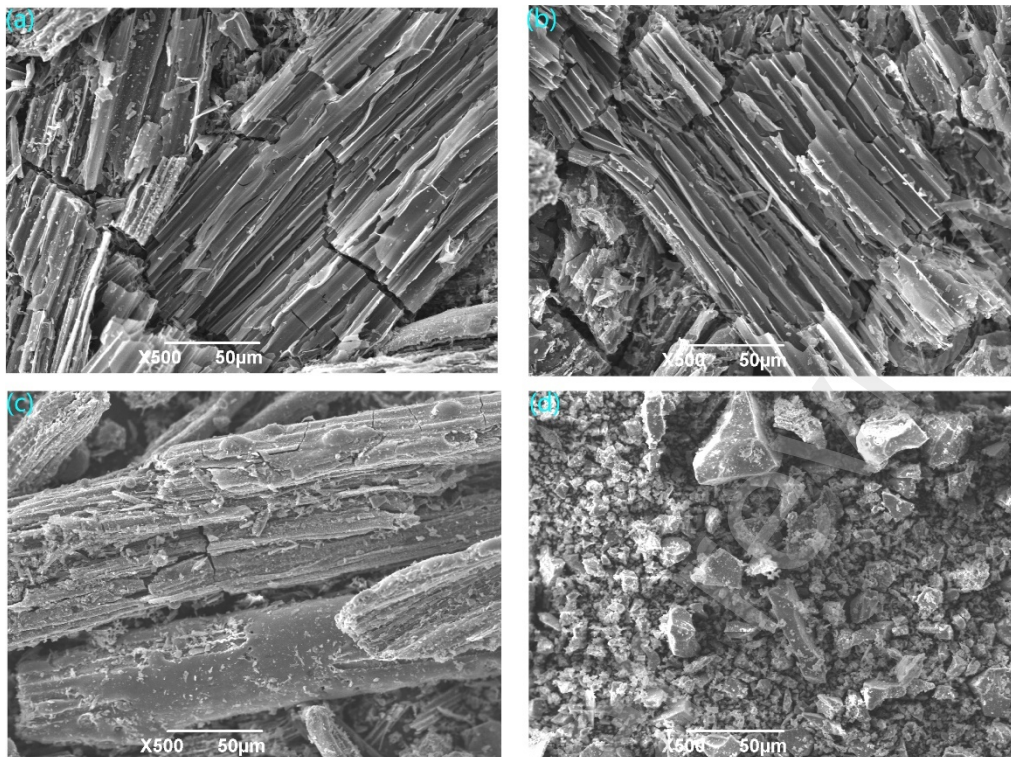


Fig. 4 The SEM images delineate the progressive structural changes in straw residues at escalating temperatures of 600°C (a), 700°C (b), 800°C (c), and 900°C (d) during gasification.

3.2. Characteristics of RDF gasification

In the pursuit of sustainable waste management, optimizing the S/M ratio is critical for enhancing resource recovery through steam gasification while mitigating the environmental impact. A lower S/M ratio exhibited higher concentrations of CO and H₂, as indicated in Fig. 5a, which is desirable for syngas production. However, as shown in Fig. 5b, lower S/M ratios increased gas yield. Conversely, a higher S/M ratio demonstrated reduced tar production and an increase in tar conversion efficiency, aligning with the goal of waste-harmless disposal by minimizing the output of harmful byproducts. Moreover, the inflection points in Fig. 5c suggest a threshold beyond which additional steam does not significantly enhance the gasification process but affects the thermal efficiency, as corroborated by the reduction in cold gas efficiency at higher S/M ratios.

In Fig. 5a, we can observe the influence of the S/M ratio on the synthesis gas component yields. Higher ratios appear to increase the yield of H₂ and CO, which is favorable for syngas applications such as energy recovery and chemical synthesis, due to their high combustion energy and chemical reactivity, respectively. This is consistent with the promotion of the steam reforming and water-gas

shift reactions at elevated steam concentrations, leading to enhanced H_2 production. Conversely, CO_2 and light hydrocarbon gases like CH_4 and C_2H_4 display a decrease in yield with higher S/M ratios. The reduction in CO_2 yield could possibly be explained by a shift in the Boudouard reaction towards the production of CO, while the suppression of CH_4 and C_2H_4 suggests the advancement of hydrocarbon cracking and reforming pathways with an increase in steam supply. The impact of the S/M ratio on the syngas yield and its lower heating value is presented in Fig. 5b. The syngas yield shows a negative correlation with the S/M ratio, indicating that past a certain point, additional steam does not contribute to increasing the yield, likely due to the dilution effect or energy losses from the endothermic steam gasification reaction. This is accompanied by a reduction in LHV, likely reflecting an increase in non-combustible gas fractions and a diluted energy content of the syngas.

Fig. 5c offers insights into the process efficiency, illustrated by the changes in carbon conversion and cold gas efficiency across different S/M ratios. Both metrics exhibit pronounced peaks at intermediate S/M ratios, after which the efficiency decreases, suggesting an optimal S/M ratio for achieving maximum process efficiency. This optimal point represents a balance between sufficient steam availability for promoting desirable gasification reactions and the avoidance of energy losses due to the excess steam acting as a thermal ballast, diluting the syngas. Finally, Fig. 5d elucidates the variation in the relative distribution of tar, solid, and gaseous products. As the S/M ratio increases, tar production significantly decreases, likely due to enhanced tar reforming reactions in the presence of steam. The consistent production of solid byproducts across varying S/M ratios suggests that steam does not significantly affect the yield of solid residues, while the proportion of gas products largely increases, emphasizing the role of steam in driving the gasification reaction towards a more complete conversion of solid waste into gaseous fuels. The interplay between the S/M ratio and these different gasification outcomes highlights the complexity of optimizing RDF gasification processes. It underscores the need for a careful balance to maximize syngas quality and minimize byproducts while ensuring high carbon conversion and process efficiency.

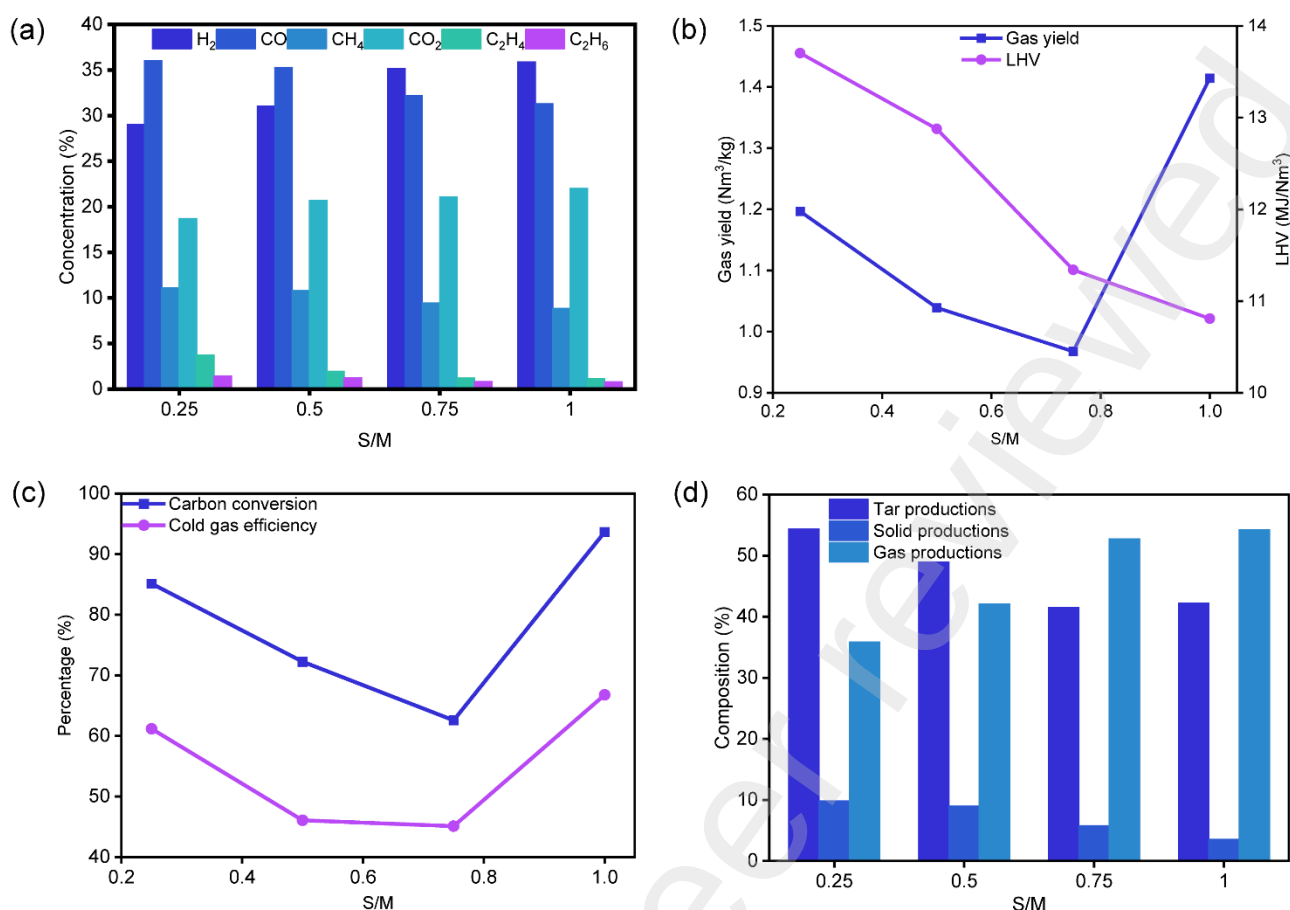


Fig. 5 Effects of S/M on various parameters relevant to the steam gasification of RDF: (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions compositions.

Fig. 6a shows the impact of temperature on the composition of syngas produced from RDF gasification. The positive correlation between temperature and the yields of H₂ and CO can be attributed to enhanced endothermic reactions such as steam reforming and partial oxidation at higher temperatures. This results in the thermal cracking of organic macromolecules within the waste, producing a richer hydrogen content, as higher temperatures favor the disassociation of water molecules in the steam reforming process. Additionally, the decrease in CH₄ and C₂H₄ with increasing temperature may indicate secondary reactions where these species are consumed to generate more H₂ and CO. Fig. 6b presents the influence of temperature on both the syngas production rate and its LHV. There is an apparent enhancement in the production rate with temperature, suggesting that the gasification reactions become more effective at higher thermal conditions. Contrarily, the LHV of the produced syngas increases modestly with temperature, potentially due to the improved quality of the syngas with a higher ratio of combustible gases (H₂ and CO) over incombustible ones (CO₂ and

steam). These observations support the notion that the thermal efficiency of syngas production is optimized at elevated temperatures, which is vital for energy recovery applications.

The variation of carbon conversion and cold gas efficiency with temperature in Fig. 6c further accentuates the importance of thermal conditions in optimizing gasification performance. Carbon conversion efficiency increases with temperature, signifying that a higher fraction of the RDF's carbonaceous content is effectively transformed into gaseous products. This result is concurrent with the rising trend of cold gas efficiency, which is a direct measure of the chemical energy of the syngas compared to the energy input from the RDF. A higher efficiency indicates a more effective conversion of waste to energy. Fig. 6d provides insight into the temperature influence on the relative distribution of tar, solid, and gaseous products. Higher temperatures considerably reduce tar formation, likely due to thermal cracking processes that break down complex organic compounds. The ratio of solid byproducts decreases as well, while the gas fraction augments, indicating a shift towards more complete gasification with temperature escalation.

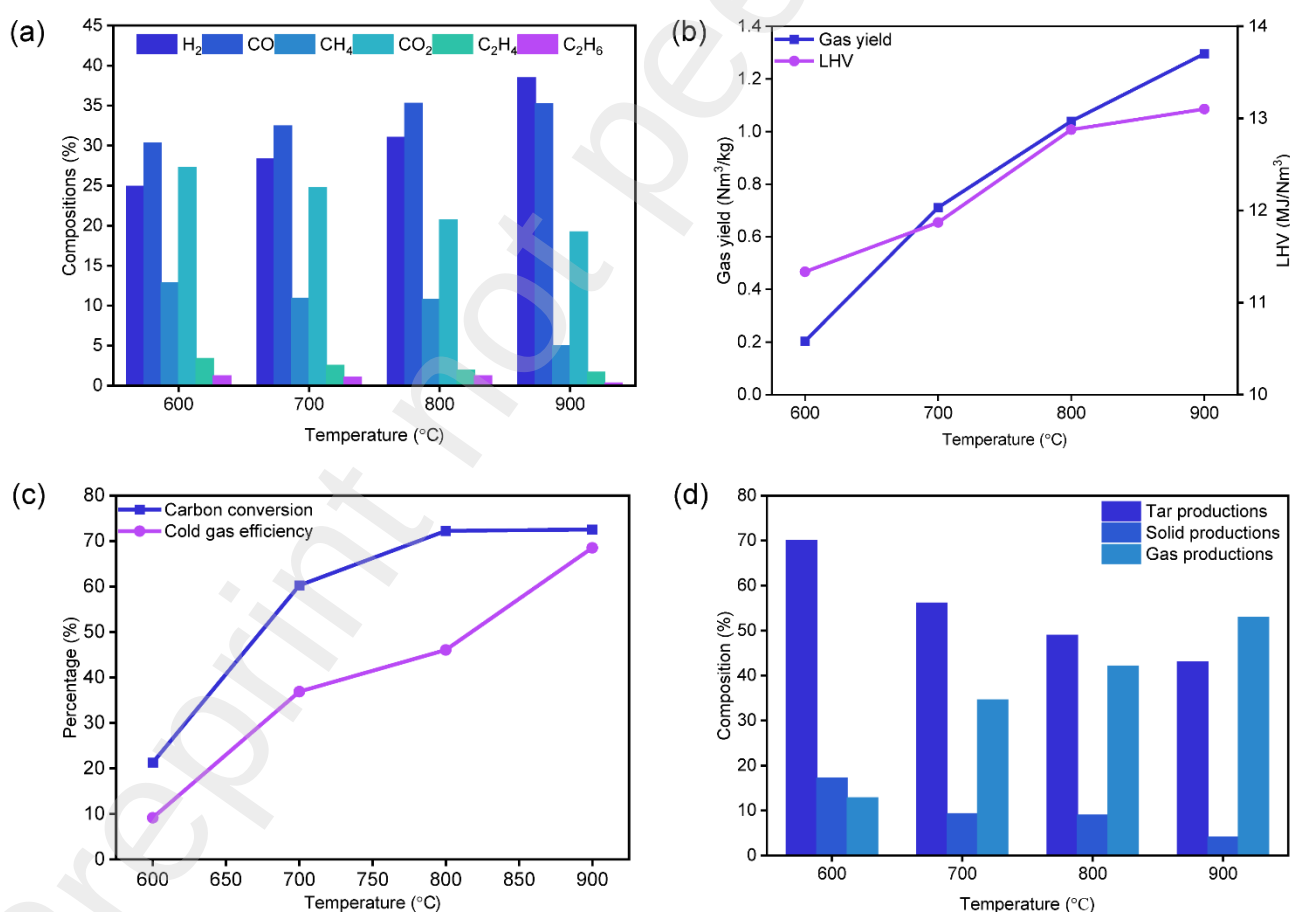


Fig. 6 Effects of temperature on various parameters relevant to the steam gasification of RDF: (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions

compositions.

Fig. 7 shows the evolutionary behaviors of the EDS data of the bottom RDF ash for different temperatures at an ER of 0.2. The characterization of solid residues following RDF steam gasification is crucial for understanding the thermal transformation properties and for developing efficient waste management practices. In Fig. 7a, the X-ray diffraction (XRD) patterns of the solid by-products at different temperatures revealed that the crystallinity and phase composition of the residues evolves with temperature. Specifically, peaks corresponding to salts such as KCl and CaCO_3 diminish with increasing temperature, indicating thermal decomposition and potential volatilization of these components. At higher temperatures (e.g., $T=900^\circ\text{C}$), the emergence of new phases, like Ca_2SiO_4 , suggests a reformation of inorganic constituents, which may have implications for the safe disposal and stabilization of the remaining ash. Coupled with the elemental composition analysis in Fig. 7b, the trends observed underscore significant temperature-driven changes in the distribution of major elements within the solid residue. For instance, the concentrations of volatile elements such as Na and K decrease with temperature, corroborating findings by Deng et al. [20] on the volatility of alkali metals during gasification. The retention of heavy metals, such as Ca and Fe, presents opportunities for resource recovery while necessitating strategies for the containment of such elements to prevent leaching and environmental contamination.

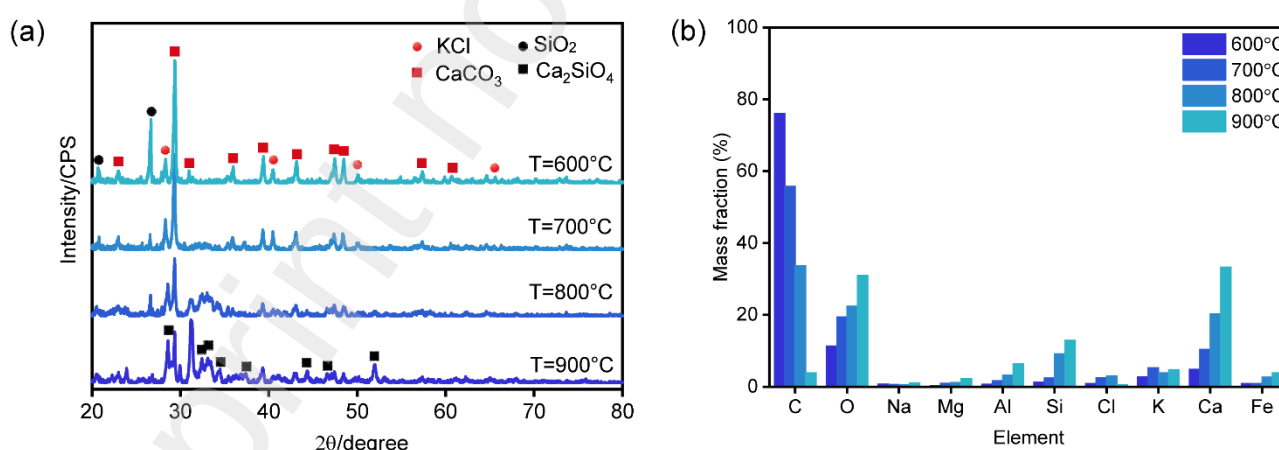


Fig. 7. (a) the XRD patterns of the solid by-products from RDF at different temperatures; (b) the selected area in the SEM images of the bottom RDF ash for EDS analysis with an S/M of 1.0 at the different temperature.

SEM analysis of solid residues generated from RDF steam gasification reveals the impact of temperature on their microstructural properties. The SEM images at distinct temperatures shown in

Fig. 8, illustrate a clear morphological transition from relatively intact fibrous structures to a more amorphous and agglomerated state. This result indicated that increasing temperature during gasification leads to enhanced thermal decomposition of organic components and facilitates the agglomeration of inorganic matter.

At the lower temperature of 600°C, the SEM image suggests preservation of the original textural features of RDF, with fibers recognizable, reflecting limited thermal degradation (Fig. 8a). As the temperature increased to 700°C (Fig. 8b) and 800°C (Fig. 8c), progressive morphological changes manifest, characterized by significant deformation and partial melting of the inorganic constituents, indicative of the onset of ash sintering. Finally, at 900°C (Fig. 8d), the SEM analysis demonstrates a nearly complete morphological transformation into a glass-like structure, pointing towards a substantial viscous flow and sintering of the inorganic materials. These microstructural changes have implications for both the resource recovery potential and the environmental impact of the gasification process. The increased sintering at higher temperatures could result in decreased porosity and increased leachability of toxic elements from the solid residues, suggesting more rigorous post-treatment procedures may be necessary for safe disposal. The SEM characterizations provide invaluable insight into optimizing gasification parameters to maximize energy recovery from RDF while minimizing potential environmental hazards of the resulting solid by-products.

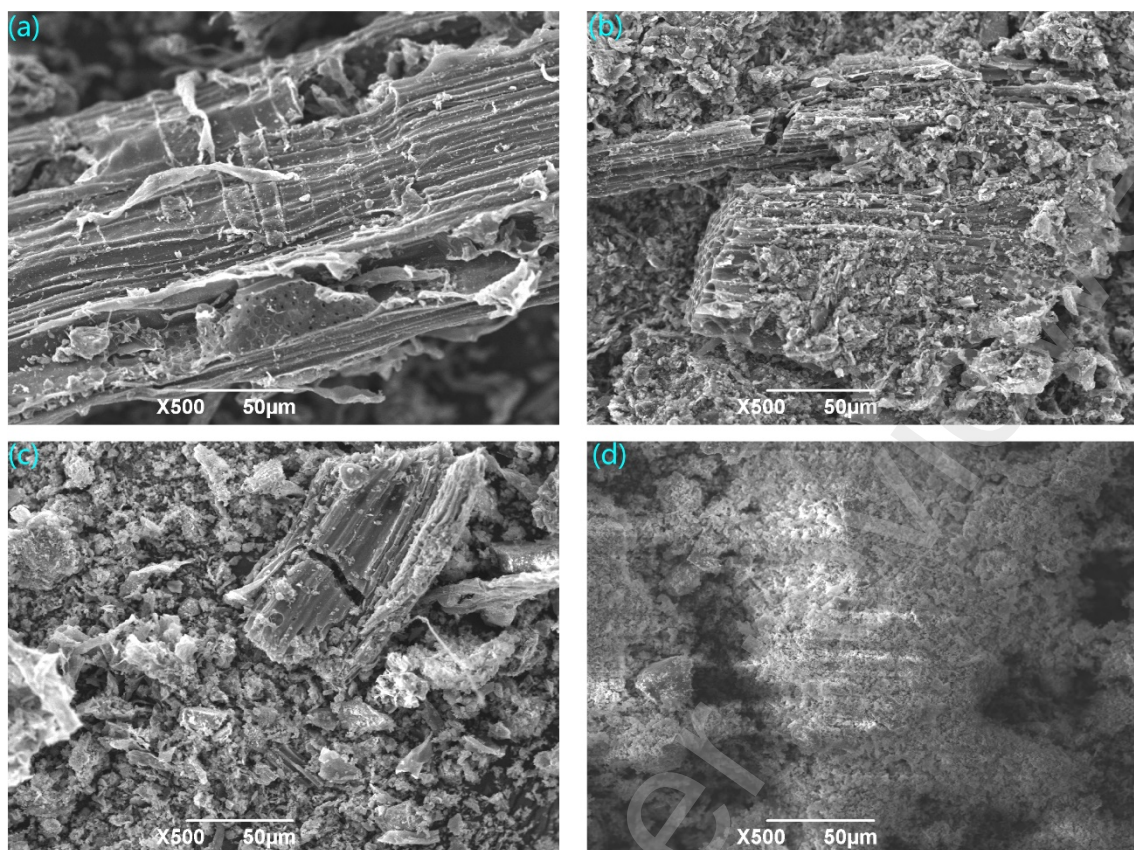


Fig. 8. The SEM images delineate the progressive structural changes in RDF residues at escalating temperatures of 600°C (a), 700°C (b), 800°C (c), and 900°C (d) during gasification.

3.3. Characteristics of the co-gasification of RDF and straw

Fig. 9a graphically illustrates how the syngas composition depends on the S/M ratio, evidencing a general increase in the generation of H_2 and CO , the two major energy carriers in syngas, with higher S/M ratios. This trend is attributable to the improved steam reforming reaction kinetics and intensified water-gas shift reactions at elevated steam inputs. The decrease in the production of CO_2 and CH_4 with increasing S/M ratio hints at the shifting of reaction equilibria towards the production of H_2 and CO , which positions the gas output as a more efficient energy source. The variations in the levels of CH_4 and higher hydrocarbons like C_2H_4 further suggest enhanced cracking of heavier hydrocarbons at higher steam inputs, optimizing gas yield for subsequent energy applications. Fig. 9b, on the other hand, reveals a non-linear relation between the S/M ratio and both the syngas yield and LHV. The syngas yield peaks at a certain S/M ratio, beyond which it sees a downturn, underpinning the concept of an optimum steam input for the gasification process. The decline in LHV alongside increasing S/M ratios manifests the dilution effect excessive steam has within the gas phase, lowering the combustible gas concentration and thus the overall energy content available for

exploitation.

Furthermore, Fig. 9c elucidates the influence of the S/M ratio on carbon conversion and cold gas efficiency. An ideal S/M ratio is designated by the apex of the plots, which corresponds to a maximization in the carbon conversion-reflecting the successful transformation of the carbonaceous material in the waste mix into a gas efficiency-an indicator of the process's efficacy in turning feedstock into valuable energy. Lastly, as demonstrated in Fig. 9d, adjusting the S/M ratio significantly affects the ratio of tar to solid and gaseous products, with a remarkable decrease in undesirable tar production at high S/M ratios. Consequently, elevating the S/M ratio proves advantageous in augmenting the purity and quality of the gas products, widening their potential for energy recovery applications.

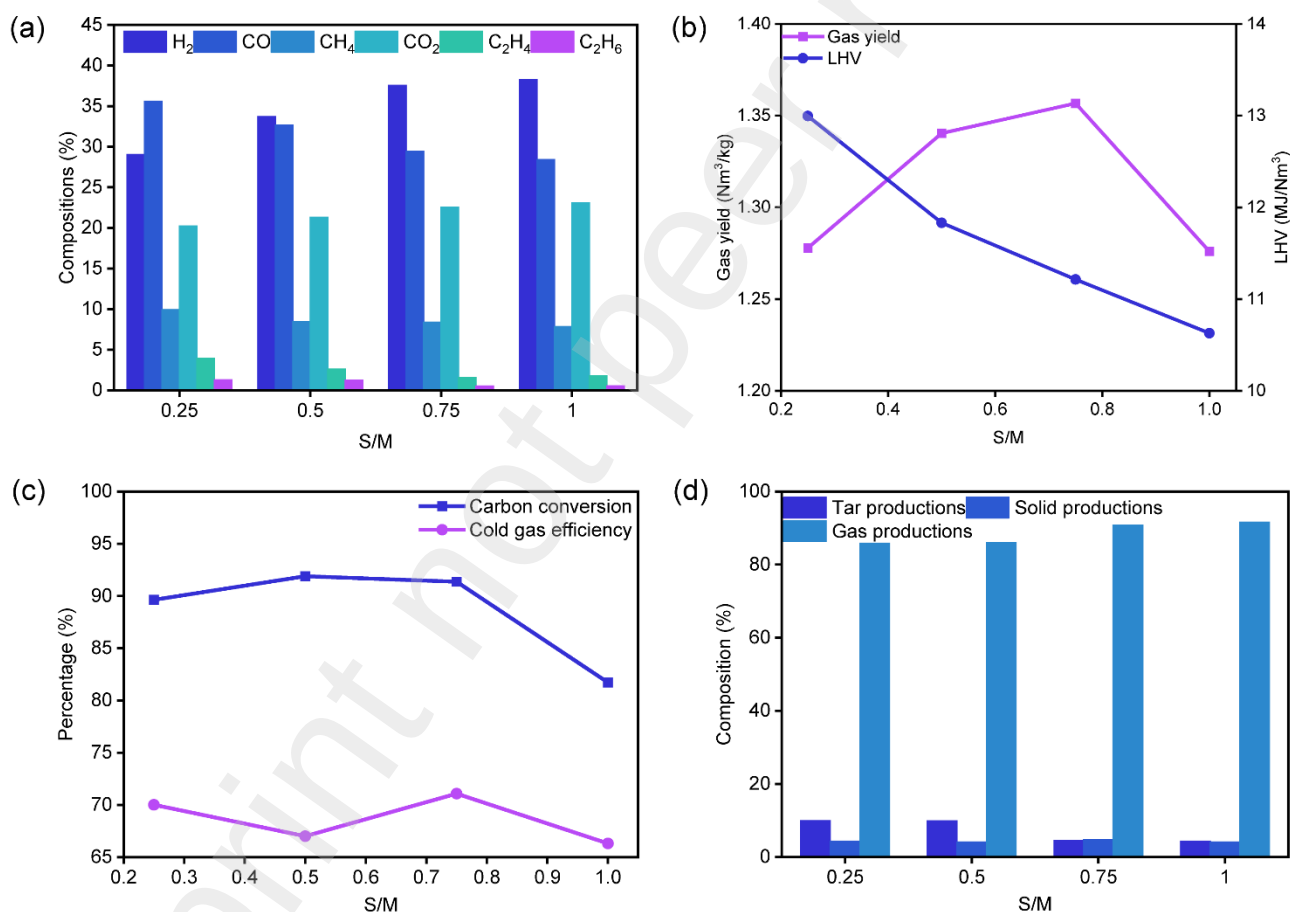


Fig. 9 Effects of S/M on various parameters relevant to the co-steam gasification of RDF and straw (mass ratio of RDF and straw is 1, gasification temperature is 800°C): (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions compositions.

Fig. 10a presents the response of syngas components to varying temperatures during the gasification process. The graph indicates a substantial increase in H₂ and CO yields at higher

temperatures, signifying improved thermal decomposition and gas reforming reactions. Enhancement of these components at elevated temperatures can be linked to the effective breakdown of organic compounds within the waste mixture, underlining the endothermic nature of steam gasification reactions which mobilize carbon into energy-rich gaseous forms. Conversely, the diminishing yields of CO₂ and hydrocarbon gases such as CH₄ suggest a favorable shift towards the production of more valuable syngas constituents with higher calorific potential. In Fig. 10b, the syngas yield and LHV trends indicate that the production rate of syngas climbs with temperature, suggesting that the higher energy input facilitates more efficient waste conversion. The accompanying rise in LHV with temperature confirms that such syngas have a greater energetic concentration, underscoring the potential for enhanced energy recovery from the bio-waste mixture. This result demonstrates the necessity of optimizing the temperature to maximize the output of calorific gases for energy applications.

Fig. 10c expounds on the effects of temperature on both carbon conversion efficiency and cold gas efficiency, pivotal metrics in judging the performance of gasification. Increment in temperature correlates with higher carbon conversion efficiency, indicating a more complete metamorphosis of the feedstock's carbon into gaseous fuels. The cold gas efficiency, which represents the energetic efficiency of converting feedstock into usable syngas, displays a similarly positive relationship with temperature, reflecting the increased quality of energy obtainable from the waste by thermochemical means. Fig. 10d elucidates the distribution of byproducts - tar, solids, and gas outputs. Higher operational temperatures substantially reduce tar formation, which is beneficial for the gasifier's operation and syngas cleansing steps. In contrast, the increase in the gas fraction at higher temperatures promotes the overall goal of energy recovery, while solid residues decrease, signifying a more comprehensive waste-to-energy conversion.

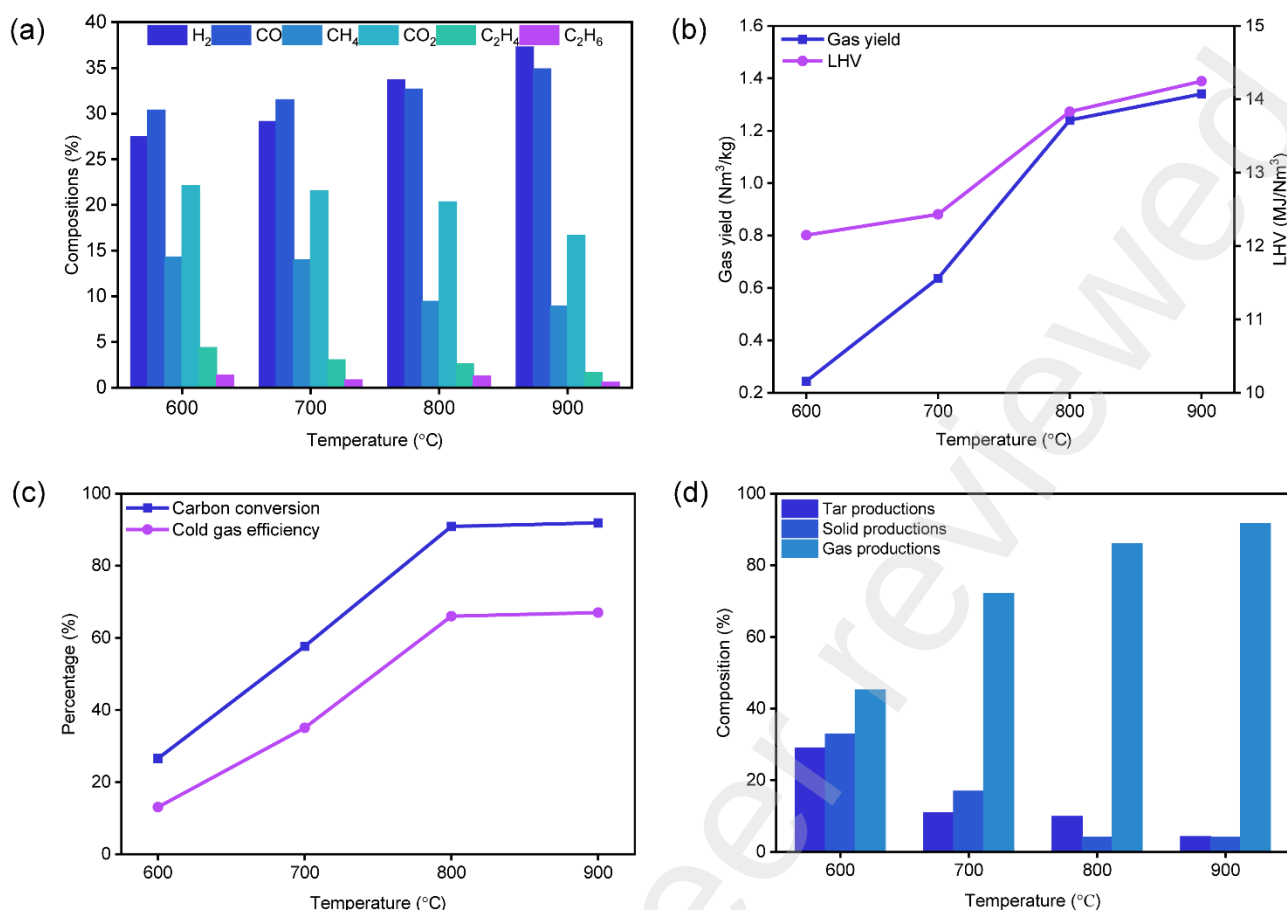


Fig. 10. Effects of temperature on various parameters relevant to the co-steam gasification of RDF and straw (mass ratio of RDF and straw is 1, gasification temperature is 800 °C): (a) gas composition, (b) gas yield and LHV, (c) carbon conversion efficiency and cold gas efficiency, (d) total tar/solid/gas productions compositions.

The XRD patterns displayed in Fig. 11a indicate the presence of inorganic phases such as KCl, SiO₂, and K₂SO₄ across a range of gasification temperatures, with the relative intensities of diffraction peaks providing insight into the stability and transformations of these compounds. Notably, peaks corresponding to calcium compounds such as CaCO₃ can be seen altering in height across the temperature spectrum, which may be indicative of calcination or other decomposition reactions. Such thermal transformations are important regarding the valorization or immobilization of ash residues, as they affect the leachability and physical properties of these waste-derived materials.

Fig. 11b complements the XRD findings by providing elemental composition information through EDS. The variations in elemental abundance, particularly of catalytically and structurally significant elements like Al, Si, K, and Ca, across different temperatures, suggest a redistribution and potential chemical modification of these constituents. The thermal behavior of these elements, such as volatilization or formation of new compounds, can influence the residue's properties and determine

its potential use as a secondary material in applications like construction or agriculture.

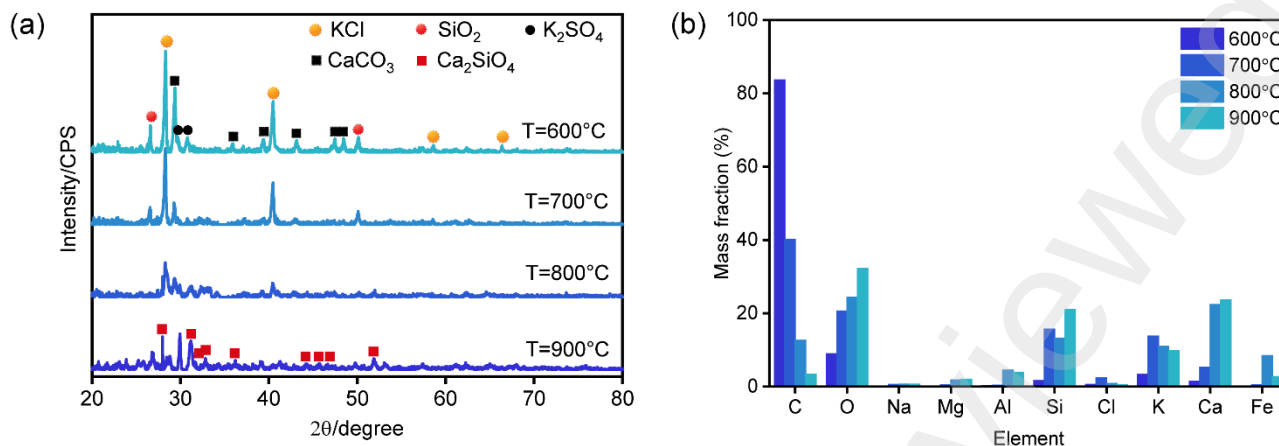


Fig. 11. (a) the X-ray diffraction (XRD) patterns of the solid by-products at different temperatures; (b) the selected area in the scanning electron microscope (SEM) images of the bottom refuse derived fuel (RDF) ash for energy-dispersal spectrometer (EDS) analysis with an S/M of 1.0 at the different temperature.

Observations from Fig. 12a show the solid residue at 600°C with well-preserved lamellar structures and fibrous integrity, indicating a lower degree of thermal degradation. The stratified layers are characteristic of cellulose and hemicellulose, components of straw, that have undergone initial thermal attack but retain much of their original structure. Elevating the temperature to 700°C, as shown in Fig. 12b, leads to a noticeable change in the char morphology. The appearance of more fragmented and less defined structures suggests a higher degree of decomposition. The fibrous elements show signs of disintegration, which correlates with the ongoing devolatilization and conversion of the organic matter. This intermediate state reflects a transition where increased porosity may enhance the accessibility of reactive sites, thereby facilitating further gasification reactions.

At 800°C and 900°C, Fig. 12c and Fig. 12d respectively, there is a clear progression towards a more granular and amorphous char structure. The thermal decomposition is so advanced that the stratification is no longer visible, and the porosity has increased significantly. At the highest temperature of 900°C, the residue exhibits extensive sintering with glassy surface morphology, indicating the near completion of organic matter removal and the formation of largely inorganic ash. Higher temperatures promote the agglomeration and fusion of ash particles, creating large voids and a potentially vitrified surface, which implies a substantial alteration in the chemical and physical properties of the residues. These morphological changes indicate that as the temperature increases,

the efficiency of conversion of the waste materials into syngas improves, reducing the quantity and toxic potential of the residual solid waste. This evolution is critical for understanding the optimization of waste-to-energy processes and enhancing the inertness of the by-products, contributing to the goal of harm reduction in waste management.

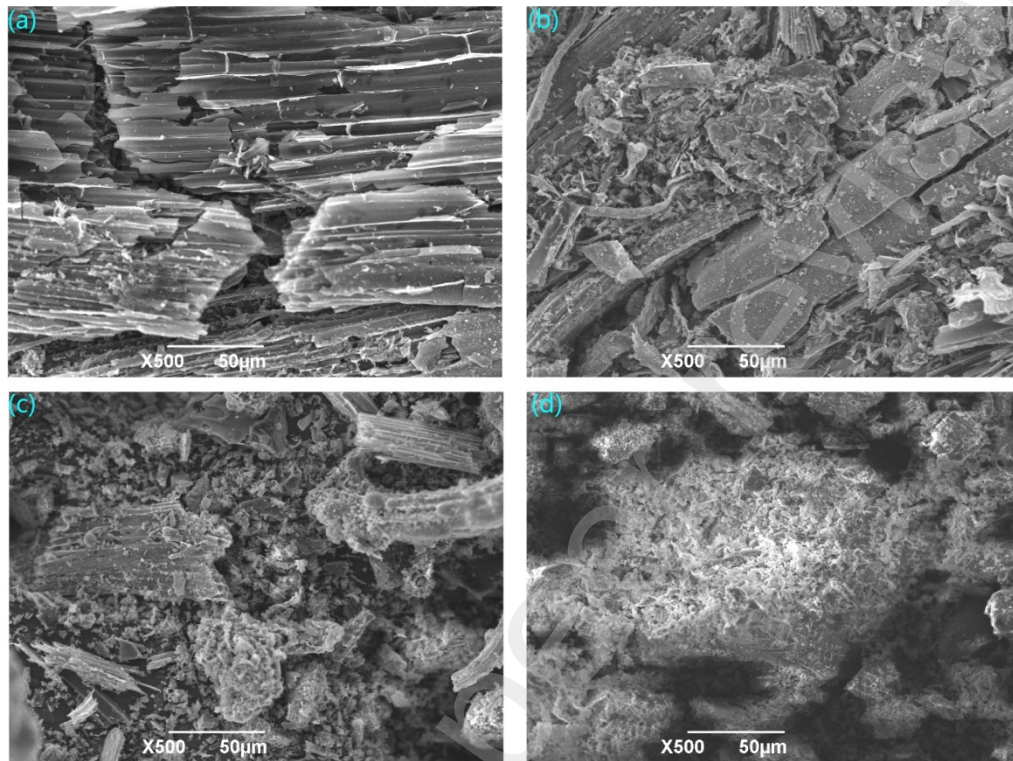


Fig. 12. The SEM images delineate the progressive structural changes in RDF and straw residues at escalating temperatures of 600°C (a), 700°C (b), 800°C (c), and 900°C (d) during gasification.

3.4 Synergistic effects from the co-gasification of RDF and straw

To quantify the synergistic effects of MSW and straw during gasification, the synergistic effects were calculated using the following formula: $\Delta M = X_c M_c + X_b M_b$, where X_c and X_b represent the mass fractions of the feedstocks with types straw and RDF, respectively, and M_c and M_b represent the values of gas yield and carbon conversion efficiency of types straw and RDF, respectively. The synergistic effects based on percentage were calculated using the following formula: $SEBP = (M_a - \Delta M) / M_a \times 100$, where ΔM and M_a represent the synergistic effects and the tested values, respectively.

In the realm of thermochemical conversion technologies, the steam gasification of municipal solid waste (MSW) has emerged as a pivotal process in sustainable waste management and renewable energy production. The integration of biomass as a co-gasification material with MSW can introduce substantial synergistic effects, as evidenced by the experimental outcomes encapsulated in the

presented data. These outcomes demonstrate enhanced syngas yields, carbon conversion efficiency, and cold gas efficiency at elevated temperatures. These findings bear significant implications for the advancement of gasification technology, with specific application value in the context of enhanced resource recovery and environmental sustainability.

Within the scope of process intensification, the observed synergistic gas yield (Gy) at varying temperatures and S/M ratios elucidates the alterations in reaction dynamics, potentially attributed to the catalytic interactions between the alkali and alkaline earth metallic species inherent to biomass and the various organic and inorganic constituents of MSW. These interactions may lower the activation energy for key gasification reactions, thereby facilitating an augmented production of syngas that surpasses the outputs predicted by classical equilibrium models predicted based on independent behavior of the feedstock constituents.

The application value of the increased syngas yield within an industrial context is twofold; firstly, it leads to an elevated generation of a valuable chemical feedstock for the synthesis of liquid fuels and chemicals via subsequent Fischer-Tropsch synthesis or methanol production pathways., Secondly, it provides a higher energy content gas stream compatible for direct utilization in internal combustion engines or turbines for electricity generation, thereby contributing to the displacement of fossil fuel-derived energy sources.

Carbon conversion efficiency is a vital metric in evaluating the effectiveness of waste-to-energy conversion processes. The increased CC rates underpin the catalytic effect of the ash content within the biomass, which may act to facilitate the gasification of recalcitrant carbon structures present in MSW. This higher conversion rate denotes a more complete utilization of the carbonaceous materials within the waste stream, minimizing the production of undesirable by-products such as tars and chars that can compromise process efficiency and generate disposal challenges. From a lifecycle perspective, the near-complete carbon conversion minimizes the carbon footprint of the waste valorization process, making it an attractive proposition for municipalities striving towards zero-waste initiatives.

Moreover, the positive deviation in CGE from theoretical expectations underscores the high-quality nature of the syngas produced, which is enriched with hydrogen and carbon monoxide, while being low in diluents such as nitrogen, thereby presenting a high heating value. The elevated CGE implies enhanced thermal efficiency of the gasification system and signifies a potential reduction in

ancillary fuel requirements, thereby bolstering the economic case for the commercial-scale implementation of waste gasification systems.

In conclusion, the synergistic effects manifesting in the co-gasification of MSW with biomass underpin the significant potential for this process in terms of improved efficiency, environmental compatibility, and economic viability. The enhanced performance parameters observed in experimental settings highlight the need for the optimization of process conditions and feedstock compositions to realize the full potential of this technology. The deployment of such optimized gasification systems can significantly contribute to the development of a sustainable waste management infrastructure that aligns with contemporary energy policy objectives aimed at reducing reliance on non-renewable energy sources and mitigating the environmental impact of waste disposal.

Table1 Synergistic effects from the co-steam gasification of refuse derived fuel (RDF) and straw

S/M	Synergistic Gy/%				Synergistic CC/%				Synergistic CGE/%			
	600°C	700°C	800°C	900°C	600°C	700°C	800°C	900°C	600°C	700°C	800°C	900°C
0.1	23.88	-21.49	-15.21	-5.09	25.75	-20.37	-10.13	-3.94	41.41	-3.07	-0.75	-1.88
0.2	23.58	-4.87	0.85	-0.22	8.55	-7.15	5.10	11.41	26.14	1.94	22.72	13.72
0.3	14.60	11.92	14.43	8.92	16.16	11.96	12.07	6.15	24.78	26.80	26.42	15.98
0.4	2.31	13.78	8.27	2.6	5.24	16.29	13.23	-2.98	16.02	31.11	12.17	12.67

4. Conclusions

In this investigation, the aim was to assess the individual and combined effects of gasification temperature and S/M ratio on the co-steam gasification of biomass and RDF. These experiments confirmed that the optimal S/M ratios enhance syngas quality by maximizing H₂ and CO yields while reducing tar and other by-products in the steam gasification of straw. Increasing temperatures further promote syngas production, as evidenced by higher H₂ and CO levels and improved gas yields and efficiencies. Solid residue analyses confirm temperature-dependent mineral transformations and structural changes, supporting the optimization of gasification conditions for better energy recovery and environmental compliance. The optimization of the S/M ratio and gasification temperature is crucial for efficient RDF gasification. Lower S/M ratios enhance syngas (CO and H₂) production, yet

higher ratios reduce tar and increase tar conversion efficiency. The optimal S/M ratio and temperature balance syngas yield, quality (LHV), and process efficiency (carbon conversion and cold gas efficiency), while minimizing environmental hazards from solid residues. Temperature elevation promotes syngas production and quality, reduces tar and solid by-products, influencing residue crystallinity and morphology. The investigation into the co-gasification of RDF and straw reveals notable synergies that enhance syngas yield, carbon conversion efficiency, and cold gas efficiency, particularly at elevated temperatures and optimal S/M ratios. These synergistic effects are attributed to catalytic interactions between feedstock constituents, leading to a gas output that is rich in H₂ and CO, with reduced levels of CO₂ and CH₄, indicating an optimized energy source. The optimal S/M ratio and temperature conditions facilitate the maximization of carbon conversion and the efficiency of the gasification process, thereby reducing the production of undesirable by-products such as tar. Additionally, the increased gas yield and energy content of the syngas highlight the potential of this technology for sustainable waste management and renewable energy production. This process not only enhances the efficiency and environmental compatibility of waste-to-energy conversion but also supports the advancement of gasification technology for improved resource recovery and reduced environmental impact.

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